

Public Economics Practicum 7

Discussion Questions Rosen:

All questions will be discussed in the tutorial.

	8th and 9th editions	10th edition
Chapter 6	1 b), 3, 11	1 b), 3, 10

In addition to 3:

Is there anything C can do to improve upon the log-rolling outcome?

In addition to 11:

Suppose John can set the agenda. What is the outcome if people vote sincerely? What is the outcome if they vote strategically? Is having agenda-setting power beneficial or not?

Additional Exercises:

7.1 Consider the preferences as in Rosen's DQ 11b. Suppose that, rather than using pairwise majority voting, the three people use plurality voting: they vote over the three alternatives simultaneously, and each person can cast one vote. The alternative that gets the highest number of votes wins. Also, suppose that in case multiple alternatives receive the highest number of votes, John has the power to decide among these alternatives (he is the 'chairman').

- (a) Show that if they vote sincerely, the outcome is M .
- (b) Show that if they vote strategically, a Nash equilibrium exist in which M is not the outcome. Assume that if people are indifferent, they vote for their most-preferred option.
- (c) Is being the chairman beneficial or not?

7.2 5 people vote about how to spent the afternoon together. They can choose between the Swimming Pool (S), the Library (L), and the Bar (B). The preferences of John and Jim are such that $S > L > B$. Tom, Claire, and Carla prefer $B > S > L$.

- (a) If they use the majority rule and vote sincerely, what is the outcome?

Now suppose they use rank-order voting. This is a voting procedure where each individual ranks the alternatives. The highest ranked alternative gets 1 point, the second-highest 2 points, etc. The alternative which receives the lowest total number of points wins.

- (b) Compared to majority rule, what could be an advantage of using rank-order voting?
- (c) If they vote sincerely, what is the outcome?
- (d) What would be the outcome if the only alternatives were S and B ?
- (e) Relate the answers to (c) and (d) to Arrow's Impossibility Theorem.

7.3 In Utopia, life is almost perfect. The only issue on which citizens have different opinions is on the size of the beach s . The utility of citizen i from a beach of size s is

$$U_i = -|s - s_i^*|$$

where s_i^* is citizen i 's most preferred size of the beach. Among citizens, s_i^* is uniformly distributed between 0 and 1 mile.

- (a) Are these preferences single-peaked?
- (b) What is the s_i^* of the median citizen?

Utopia is a representative democracy and holds presidential elections. The elected president will decide about the size of the beach. There are two candidates, mr. Red and mrs. Brown. They have similar preferences as citizens. The ideal point of mr. Red is $s_R^* < \frac{1}{2}$, the ideal point of mrs. Brown is $s_B^* > \frac{1}{2}$.

Suppose that before the elections, a candidate can credibly promise to choose a certain beach size when elected.

- (c) If mr. Red commits to s_R and mrs. Brown to s_B , what is the probability that mr. Red wins the election?
- (d) To which beach size will the candidates commit? Who wins the election?
- (e) If the candidates would be unsure about the preferences of the median voter, what would be the trade-off they face in setting their promise about s ?

Now suppose that promises are not credible. In fact, when elected, candidates can choose any beach size, regardless of their campaign promises.

(f) Who wins the election? What will be the size of the beach?

- 7.4 Angela, David, and Emmanuel share an apartment. They have agreed to hire a cleaning company to clean their apartment. However, they still need to decide for how many hours H the apartment should be cleaned. The cleaning company charges 12 euro per hour of cleaning. Angela's valuation of cleaning is given by

$$U_A = 40H - H^2 \text{ euro.}$$

David's valuation is given by

$$U_D = 60H - \frac{3}{2}H^2 \text{ euro.}$$

Emmanuel's valuation is given by

$$U_F = 20H - \frac{1}{2}H^2 \text{ euro.}$$

- a) Show that the socially efficient level of H is 18 hours.

Suppose that they agree on sharing the cost of cleaning equally, such that each of them pays 4 euro per hour of cleaning. Furthermore, suppose that they decide on the level of H by pairwise majority voting.

- b) Show that (given the cost sharing), each individual's preferences regarding H are single-peaked.
- c) Which level of H wins when they use the pairwise majority voting procedure?

David is unhappy about the outcome of the vote on the number of hours of cleaning H . He threatens to stop contributing to the cleaning cost. After a heated debate, Angela, David, and Francois cancel their agreements regarding cleaning. As a result, they must decide independently how many hours of cleaning to buy individually from the cleaning company.

- d) In the resulting Nash-equilibrium, David pays for $H = 16$ hours of cleaning.

Answers to exercises week 7

Rosen 6.1

b) C wins in every pairwise vote. Thus, there is a stable majority outcome, despite the fact that persons 1, 2, and 3 have double-peaked preferences. This demonstrates that although multi-peaked preferences may lead to voting inconsistencies, this is not necessarily the case.

Rosen 6.3

Without vote-trading, neither bill would pass. If there is vote-trading, then voter B would agree to support issue X provided voter A supports issue Y, allowing both bills to pass. The change in net benefits is +3 for Issue X and -2 for Issue Y, so logrolling results in a gain of +1.

Addition: Logrolling is very bad for C. Hence, to prevent this, C may propose to A to vote in favour of X if A votes against Y. Compared to having both bills passed, this benefits both A and C, and increases welfare by +3.

Rosen 6.11 / 6.10

a) The outcome of the first election (M vs. H) is M. The outcome of the second election (H vs. L) is L. The outcome of the third election (L vs. M) is M. Majority rule leads to a stable outcome since M defeats both H and L. Giving one person the ability to set the agenda would not affect the outcome in this case.

b) With the change in Eleanor's preference ordering, majority rule no longer generates a stable outcome. In a vote between M and H, the outcome is H. In a vote between H and L, the outcome is L. In a vote between L and M, the outcome is M. Hence, the voting paradox arises: even though individual preferences are all transitive, the 'collective' preference ordering is not.

This implies that it matters who sets the agenda. For example, Abigail prefers H, so she might pit L against M first in order to eliminate L and avoid having L defeat H. Thus, by having a first-round vote between L and M and a second-round vote between the winner of round 1 and H will result in H being chosen.

Addition: With sincere voting, if John first sets L against H to eliminate H. Then follows L against M, such that his favorite, M, wins.

If they vote strategically, this does not work. In the first round, Eleanor understands that if she follows her preferences and votes for L, then she ends up with M, her least favorite outcome. Hence, she optimally votes for H in the first round, such that in the second round H wins (from M).

However, John can still get his favorite, M. To do so, he must make sure that in the second round, M is up against L (see b)). Starting with H versus L does not work (see above). Starting with M versus L, then the second round has either M versus H (H wins) or H versus L (L wins). As two prefer L over H, L will win. Hence, M does

not win. Lastly, by starting with M versus H, the last round has either M versus L (M wins) or H versus L (L wins). As two people prefer M over L, M will win. Hence, John optimally starts with M versus H.

It follows that agenda-setting power is beneficial, both in case of sincere and in case of strategic voting.

Additional exercise 7.1

a) They all vote for their most-favoured option, implying that all options receive the same number of votes. Hence, as chairman, John can decide. He optimally decides for M.

b) Abigail and Eleanor realise that if they vote for their first-best option, they end up with M. Hence, they can do better by voting both for H. Then, the outcome is H, as it receives the maximal number of votes.

Beyond the question: Formally, there are 5 Nash equilibria. The assumption that if people are indifferent they vote for their most-preferred option reduces the number of equilibria to 2: (i) $J \rightarrow M$, $E \rightarrow H$, $A \rightarrow H$ such that H wins and (ii) $J \rightarrow M$, $E \rightarrow L$, and $A \rightarrow M$ such that M wins. Note that it cannot be an equilibrium for both Eleanor and Abigail to vote for L, as in that case John would vote for M. As Abigail prefers to get M over L, she would then be better off for H such that M is the outcome.

c) Being the chairman is good if people vote sincerely, but can be less preferable if people vote strategically.

Additional exercise 7.2

a) Tom, Claire, and Carla prefer B over L and B over S, so B wins any pairwise majority voting.

b) It uses the whole ranking of options, rather than only two options. Hence, it takes into account that even though a majority may favour a certain option, this option may be so bad for a minority that another option is preferable for ‘society’.

c) S gets $2 + 6 = 8$ points. L gets $4 + 9 = 13$ points. B gets $6 + 3 = 9$ points. Hence, S wins.

d) Without L, S gets $2 + 6 = 8$ points. B gets $4 + 3 = 7$ points. Hence, B wins.

e) Apparently, this voting procedure does not fulfill the criterion that in any situation the ranking of options should be independent of irrelevant alternatives. Note that L is the least-preferred option under (c). Yet, if we add option L, the outcome using rank-order voting changes from B to S.

Additional exercise 7.3

a) Yes, the peak is at s_i^* .

b) The median of a uniform distribution is equal to the mean, which is $s_m^* = \frac{1}{2}$.

c) If the median prefers Red’s position over Brown’s position, then a majority

of citizens prefer Red's position, leading to a victory for Red. The same holds for Brown. The median citizen prefers Red over Brown when $|s_R - s_M^*| < |s_B - s_M^*|$ and prefers Brown over Red otherwise.

d) Drawing is very helpful here. Suppose that the optimum of Brown is closer to the median then the optimum of Red: $|s_R^* - s_M^*| > |s_B^* - s_M^*|$. Hence, if they both stick to their optimum, Brown wins. This is very bad for Red, as $s_B^* \gg s_R^*$. Hence, Red can improve upon this by choosing an $s_R < s_M^*$ such that $|s_R - s_M^*| < |s_B^* - s_M^*|$ leading to a victory for Red. Now Brown is worse off, leading Brown to set a policy closer to the median than Red, which makes Red worse off, etc. They will end up setting their policy equal to the optimum of the median voter: $s_R = s_B = s_M^*$. Hence, both have equal chance to win.

e) Under (d), getting closer to the median policy was necessary to win the elections. However, getting closer to the median hurts the candidates, as their own optimal policy is not the median's. Hence, if there is uncertainty about the real median, then the candidates will trade-off the probability of winning the election (which increases by getting closer to the expected position of the median) and the policy that they have to implement when elected (which yields higher benefit when they remain close to their own optimum). Thus, when the candidates do not perfectly know the median's position, the probability of winning becomes a continuous variable of their position. As a result, in equilibrium, the optimal policy proposals will not be equal to the median's optimal position, but rather somewhere in between the median's optimal policy and the candidate's optimal policy.

f) When campaign promises are not credible, (rational) voters will foresee that candidates will simply implement their most-preferred policy when elected. Hence, the candidate with the optimal policy closest to the median's position will win the election.

Additional exercise 7.4

a) Note that cleaning is a public good here (all individuals' utilities depend on H). Hence, $MU_A + MU_D + MU_F = 12 \rightarrow 120 - 6H = 12 \rightarrow H = 18$.

b) Each individual has (net) value of H that can be written like $U = aH - bH^2 - 4H$. Maximising U gives first-order condition $a - 2bH - 4 = 0$, so that the optimum lies at $H^* = \frac{a-4}{2b}$. Taking the derivative of the first-order condition wrt H yields second-order condition $-2b < 0$. Hence, the valuation function has a unique maximum, which implies it has only one peak. Another way of arguing this is by showing that marginal utility $MU(= a - 2bH - 4) < 0$ for $H > H^*$ and $MU > 0$ for $H < H^*$. In other words, utility is decreasing in the distance between H and H^* : the graph of utility as function of H has only one peak.

c) It follows that Angela prefers $H_A^* = 18$, David prefers $H_D^* = 18\frac{2}{3}$, and Emmanuel $H_E^* = 16$. Hence, Angela is the median voter, so that her optimum wins against any

alternative in case of pairwise majority voting. Note: that the outcome happens to be equal to the social optimum is a coincidence. Generally speaking, pairwise majority voting can lead to outcomes that can be lower or higher than the socially efficient / social-welfare maximising level.

d) The only Nash equilibrium is where Angela and Francois contribute zero, and David responds by setting $H = 16$. Note that this is Nash: If David buys $H = 16$, for both MU_A and MU_F are smaller than $MC = 12$, implying that it is indeed optimal to buy zero.